

Multimodal Agentic RAG

Retrieval-Augmented Visual Question Answering
with Vision-Language Models on Encyclopedic-VQA

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Abstract

This project investigates *Multimodal Agentic RAG*, a paradigm that combines vision-language models (VLMs) with retrieval-augmented generation (RAG) and agentic reasoning to answer knowledge-intensive visual questions. Using the Qwen2.5-VL family as the backbone and a FAISS index built on EVA-CLIP-8B embeddings over a Wikipedia-derived knowledge base, we progressively design and evaluate a multi-stage retrieval architecture on a subset of the Encyclopedic-VQA benchmark.

Through iterative experimentation on static retrieval strategies we empirically discovered that an “Entity Guessing” strategy was the most effective approach. Rather than relying on pure visual search, the VLM generates concise entity hypotheses which are fused with image features to query the database. Building upon this finding, we integrated the optimal retriever into a constrained ReAct-style agentic loop. By explicitly decoupling entity identification from passage-level filtering and introducing code-level fallbacks, our final architecture mitigates contextual noise and the model’s tendency to hallucinate answers from its parametric memory when presented with irrelevant retrieved context. This report documents the design choices, ablation studies, and scaling behaviors of our proposed architecture.

1 Introduction

Vision-language models (VLMs) possess static parametric knowledge, limiting their performance on long-tail, knowledge-intensive visual queries. Retrieval-Augmented Generation (RAG) mitigates this by fetching external evidence, while *agentic* loops (e.g., ReAct [7]) empower models to autonomously decide when and what to retrieve. Building upon recent AImageLab works [2, 3], this project develops a zero-shot Multimodal Agentic RAG pipeline. Leveraging the **Qwen2.5-VL** family [1] as the reasoning backbone and the **smolagents** framework for orchestration, we investigate how compact models can navigate an image-only Wikipedia FAISS index.

Our architecture follows a progressive refinement approach. First, we focused on optimizing the non-agentic RAG retrieval by exploring a range of different strategies. To overcome the ambiguity of purely visual search, we ultimately adopted the “Entity Guessing” strategy. Prior to searching, the VLM generates concise hypotheses about the visual subject; fusing these text guesses with image features drastically improves retrieval precision. This intuition is strongly validated by the recent *Identify-Before-Answer (IBA)* framework [4], which decouples entity-level identification from section-level ranking. Second, we integrated this retriever into a constrained ReAct loop. We demonstrate that a 3B-parameter VLM can successfully orchestrate complex workflows by combining full-document retrieval, granular passages filtering (via an external critic), and code-level fallbacks.

2 Task, Dataset, and Evaluation

We evaluate our pipeline on a 1000-sample test subset of **Encyclopedic-VQA** [5], targeting knowledge-intensive visual question answering.

Task Definition and Infrastructure. Each sample consists of an iNaturalist 2021 query image, a natural language question, and reference answers. A fundamental challenge of this dataset is that many queries are *implicit* (e.g., using deictic expressions like “this bird”), meaning the entity must be visually identified before the question can be addressed. To bridge this gap, we rely on a provided multimodal infrastructure: a **Knowledge Base** mapping ~ 2 million Wikipedia URLs to their textual sections and titles, and a **FAISS Index** containing pre-computed EVA-CLIP-8B [6] image embeddings for each entry. Our retrieval design balance visual and textual components within the shared EVA-CLIP latent space.

Evaluation Methodology. All configurations are evaluated using the official Encyclopedic-VQA script. To isolate retrieval failures from generation errors, the evaluation tracks exactly what information reaches the model. Instead of just checking if the search engine found the right document, the script verifies if the correct passage survived the Critic’s filter and was actually included in the final prompt. Based on this condition, the analysis is structured around four key metrics:

- **Score All:** The overall accuracy across the entire dataset. It is computed by combining the successful predictions from both retrieval scenarios:
$$\text{Score All} = (\text{Hit Rate} \times \text{Score} \mid \text{hit}) + ((1 - \text{Hit Rate}) \times \text{Score} \mid \text{miss}).$$
- **Hit Rate:** The percentage of test examples where both the correct document and the specific relevant section are successfully retrieved and retained in the final prompt.
- **Score | hit:** The accuracy computed exclusively on the subset where the correct evidence is successfully retrieved. This isolates the model’s reading comprehension capability and its ability to effectively exploit the provided context.
- **Score | miss:** The accuracy computed on the remaining subset where the correct text was either not retrieved or filtered out. This evaluates the system’s ability to trigger the Code-Level Fallback and answer correctly relying solely on its internal parametric knowledge.

3 Retrieval Implementation

Our setup utilizes **Qwen2.5-VL-3B-Instruct** as the answering engine, enforcing deterministic (greedy) decoding for reproducibility. To capture fine-grained visual features, the vision processor is explicitly configured to a high resolution limit of $1280 \times 28 \times 28$ patches. Embeddings are extracted using EVA-CLIP-8B vision and text encoders to align perfectly with the pre-computed FAISS index.

3.1 Experimental Exploration

Before formalizing our definitive pipeline, we ablated several static retrieval strategies. The **Plain VLM Baseline** (no external context) establishes a lower bound, relying solely on parametric knowledge and achieving an overall accuracy (**Score All**) of 17.6%.

We then explored intermediate retrieval mechanisms, briefly outlined below:

- **Image-Only Retrieval:** A standard nearest-neighbor FAISS search embedding only the query image via EVA-CLIP-8B.

- **Caption-Based Re-Ranking:** The VLM generates a descriptive image caption, which is embedded to re-rank the top-10 visual results, aiming to mitigate taxonomically incorrect retrievals.
- **Dynamic Evidence Selection:** Passing only the top-1 document as context if its score margin over the second candidate exceeds a fixed threshold.
- **Rich Labeled Context:** Prepending original titles and source markers (e.g., [Source 1]) to the retrieved sections, helping the model navigate document boundaries.

These intermediate strategies provided limited improvements, revealing that post-retrieval re-ranking often degraded accuracy. The most significant leap was instead achieved through *multimodal query fusion*.

3.2 Final Static Retrieval: Entity Guessing

Our final strategy leverages the intuition that explicit textual hypotheses about the image content lie close to the target Wikipedia documents in the shared embedding space. Developed independently, this “Entity Guessing” approach strongly aligns with the recent *Identify-Before-Answer (IBA)* framework [4], which advocates training-free entity identification prior to retrieval.

A preliminary prompt instructs Qwen2.5-VL to formulate N candidate names for the main entity. These guesses are embedded via EVA-CLIP’s text encoder and linearly combined (using an empirical text weight $\alpha = 0.3$) with the L_2 -normalized image embedding. This effectively guides the FAISS search toward the correct semantic neighborhood.

During formulation, comparing the raw string [Image + Question + Guesses] versus [Image + Guesses] alone revealed that removing the user question increased the reading accuracy (**Score | hit**) from 57.9% to 59.6% and boosted the Hit Rate. This confirms that excluding implicit questions prevents semantic noise, allowing clean entity guesses to drive a more robust retrieval.

3.3 Retrieval Module Results

Table 1 reports the results for all configurations. In this early stage, we deliberately retrieved only $k = 3$ documents to test basic strategies without overloading the 3B model, as the ReAG-Critic module was not yet integrated.

Retrieval Module Configuration	Hit Rate
RAG, image-only retrieval	23.3%
RAG, caption-based re-ranking	16.3%
RAG, caption re-ranking + dynamic top- k	24.6%
RAG, image-only + rich context	23.4%
RAG, image + question fusion	25.8%
RAG, image + guesses + question fusion	25.9%
RAG, image + guesses (Entity Guessing)	26.3%
RAG, rich context + entity guessing fusion	23.3%

Table 1: Retrieval accuracy (Hit Rate) for static configurations. Fusing pure text hypotheses with the visual query (Entity Guessing) yields the highest performance.

As shown, **Entity Guessing** (3 guesses in this phase) emerges as the optimal strategy, peaking at a 26.3% Hit Rate. This represents a substantial improvement over the baseline image-only retrieval (23.3%). Attempting to inject semantic information via raw questions (25.8%) or explicit question fusion (25.9%) introduces semantic noise. Conversely, isolating the taxonomic identity via pure guesses maximizes FAISS alignment.

4 Methodology: Agentic ReAct Architecture

While the experiments described in Section 3 were crucial for isolating and optimizing the individual components of our retrieval module, they were ultimately constrained within a fixed, single-shot pipeline. In a traditional static RAG paradigm, the reasoning model acts as a passive receiver: it is forced to condition its final answer on whatever external context is retrieved, regardless of whether that context is genuinely helpful, or if the visual evidence in the image was already sufficient to answer the query.

Following the explicit requirements of the project assignment, we transitioned from this static approach by integrating our best-performing retrieval configuration into a **ReAct-style agentic loop** [7]. By interleaving explicit reasoning traces (*Thoughts*) with actionable commands (*Actions*) and environmental feedback (*Observations*), this architecture transforms the model from a passive generator into an autonomous agent. Instead of blindly processing injected documents, the agent is empowered to deliberate on the user’s query: it can evaluate its own parametric uncertainty, and autonomously decide whether external evidence is strictly necessary.

4.1 Lesson Learned from our First Implementation

Our initial agentic implementation utilized a three-tool architecture (`assess_retrieval_need`, `retrieve_knowledge`, and an optional `refine_search` tool). However, evaluating this setup revealed significant structural limitations. First, asking a 3B-parameter model to simultaneously compose valid JSON tool calls and formulate complex multi-hop retrieval queries caused cognitive overload, resulting in queries that underperformed the static baseline. Furthermore, a major design flaw in this first implementation was statically truncating retrieved documents to only their first few sections to save context window space, which severely limited recall. The agent frequently discarded all context and hallucinated answers, these failures demonstrated that a compact 3B model requires a streamlined action space and a strict decoupling between document retrieval and relevance filtering.

4.2 Redesign and Optimized Pipeline

Following the diagnostics from the first implementation, the agentic architecture was substantially redesigned. To lighten the cognitive load on the backbone, the action space was reduced to two custom tools:

- `assess_retrieval_need`: Evaluates whether external knowledge is strictly required to answer the question. It enables the agent to decide between retrieving external evidence or answering directly from its internal parametric knowledge when the visual evidence is sufficient. This tool is enforced as the first action of the episode.
- `retrieve_knowledge`: Executes the core multimodal retrieval pipeline. It generates entity guesses, combines them with the image embedding to query the FAISS index, and automatically filters the retrieved text passages via an external Critic before returning them to the agent’s memory.

The final architecture pipeline is visually summarized in Figure 1.

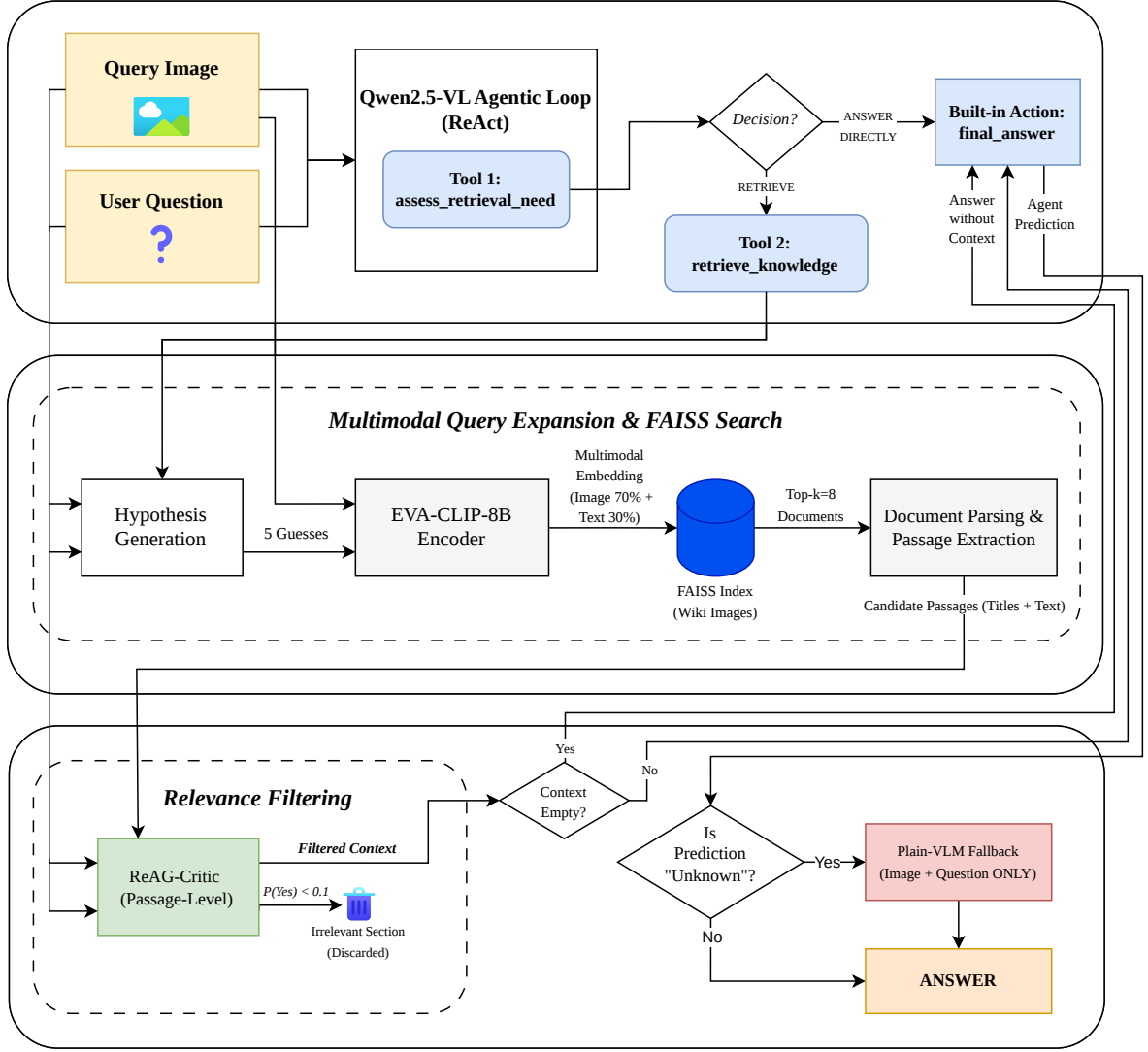


Figure 1: Overview of our final Agentic RAG architecture. The pipeline explicitly decouples multimodal query expansion from relevance evaluation (ReAG-Critic), while incorporating code-level fallbacks to handle empty contexts and prevent backbone overconfidence.

4.2.1 Full-Document Retrieval with Passage-Level Critic

To prevent the reasoning agent from being overwhelmed by lengthy texts without sacrificing recall, the `retrieve_knowledge` tool now extracts *all* passages of the top- k retrieved documents. Instead of statically truncating them, we segment the documents into individual logical passages.

Each passage is evaluated independently by **ReAG-Critic**, a specialized relevance filtering model introduced by Compagnoni et al. [3] in their Reasoning-Augmented Generation framework. The Critic receives a multimodal prompt (original image, user question, and text snippet) and outputs a binary relevance judgment. To ensure the Critic captures fine-grained visual details, its vision encoder is explicitly forced to process images at high resolution (up to $1280 \times 28 \times 28$ patches), matching the main agent’s visual acuity.

The Critic module extracts the logits from the model’s final layer and compute the softmax probability of the “Yes” token. If this probability exceeds a conservative threshold (0.1), the

passage is retained. This granular filtering ensures the agent only reads a highly concentrated, relevant context (typically 2-3 paragraphs), effectively separating *how much is retrieved* from *how much is read*.

4.2.2 Final Pipeline Summary and Separation of Concerns

To maximize precision and prevent hallucination loops, our final architecture enforces a strict *separation of concerns* across three distinct stages:

- **Embedding for FAISS:** The vector search fuses the visual features (70%) exclusively with the clean entity hypotheses (30%). By omitting the raw user question, we isolate the semantic identity of the object and maximize alignment with the image-only index.
- **Prompting for ReAG-Critic:** The filtering model evaluates the image, the retrieved passages, and the *full user question*, but remains completely blind to the generated entity guesses. This ensures the Critic acts as an impartial judge, unaffected by potentially hallucinated hypotheses.
- **Code-Level Fallback:** A programmatic safety net is triggered when the agent’s generated prediction matches specific "bailout" patterns (e.g., “Unknown”). In this scenario, the agentic loop is aborted and the final answer is generated via a clean, zero-shot inference utilizing solely the plain VLM and the original image, preventing hallucinations from misleading contexts. In our final evaluation, this mechanism was successfully triggered in **52 out of 1000** test queries.

5 Ablation Studies and Hyperparameter Sensitivity

To justify our architectural choices within the agentic pipeline, we performed three ablation studies focusing on the hyperparameters that control the flow of information between the vector store and the reasoning backbone: top- k retrieved documents, text weight (α), and the number of generated entity guesses.

5.1 Sensitivity to Top- k and Text Weight (α)

To isolate the effect of k , we ran a pure-retrieval diagnostic (using image-only queries, $\alpha = 0.0$) tracking both the Aggregate Score and the reading precision (*Score* | *Hit*). Concurrently, to find the optimal fusion alignment with the image-only FAISS index, we conducted an ablation study over the textual hypothesis weight $\alpha \in \{0.0, 0.1, 0.3, 0.5, 0.7\}$ at $k = 8$.

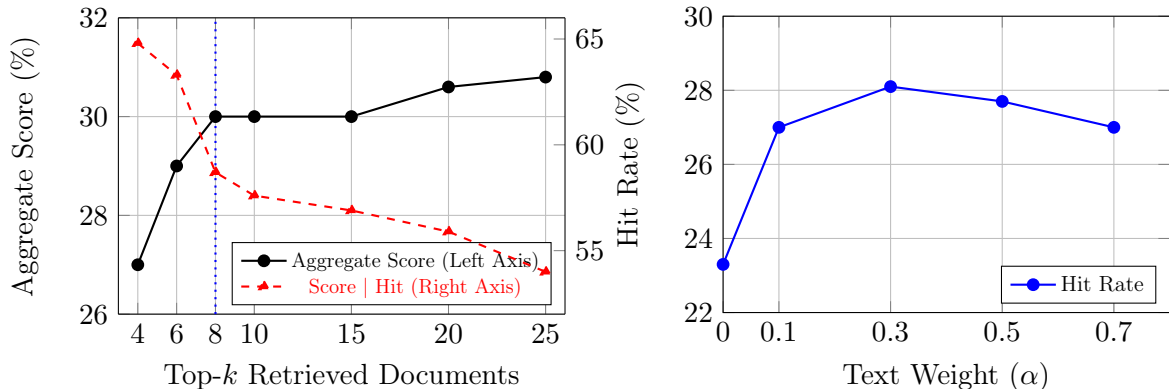


Figure 2: **Left:** Retrieval sensitivity to top- k . Retrieving excessive documents causes the reading precision (*Score* | *Hit*, dashed red line) to collapse, neutralizing aggregate gains. **Right:** Sensitivity to the textual weight α during multimodal fusion. The curve demonstrates a clear optimal sweet spot at $\alpha = 0.3$.

As illustrated in Figure 2 (Left) retrieving too many documents is inefficient. Although increasing k from 4 to 8 significantly increases the aggregate score (solid black line), extending the retrieval to $k = 15$ or beyond does not produce real improvement, yielding only fractional gains at the cost of exponentially higher inference latency. Crucially, the dashed red line (**Score** | **Hit**) exposes the qualitative cost of this scaling: as the volume of candidate sections expands, the Critic inevitably admits more False Positive passages. When the context becomes too long, the agent gets confused by irrelevant information. As a result, its reading precision drops from 64.8% ($k = 4$) down to 54.0% ($k = 25$). Consequently, we established $k = 8$ as the best balance point.

Furthermore, Figure 2 (Right) demonstrates that relying solely on visual features ($\alpha = 0.0$) bounds the hit rate at 23.3%. Introducing text hypotheses yields an immediate surge in accuracy. However, pushing α too high ($\alpha \geq 0.5$) shifts the combined vector away, causing a misalignment with the image-only FAISS embeddings and steadily degrading the hit rate. The value $\alpha = 0.3$ emerged as the empirical optimum (28.3% hit rate), providing sufficient semantic guidance without overriding the visual similarity search.

5.2 Sensitivity to Number of Guesses

A core innovation of our pipeline is the Entity Guessing mechanism. To quantify its impact and determine the optimal number of entity hypotheses to fuse with the image, we conducted an ablation study requesting $\{1, 3, 5, 10, 15, 20\}$ guesses. To isolate the retrieval performance from the stochastic variance of the agent’s main loop, we fixed a global seed and forced a near-deterministic generation temperature ($\tau = 0.1$).

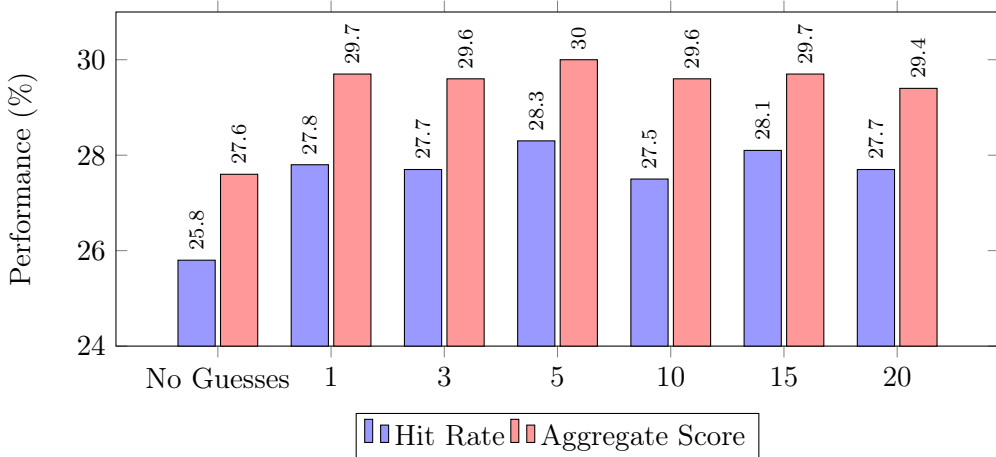


Figure 3: Impact of Entity Guessing mechanism. Introducing even a single guess vastly outperforms the “No Guesses” baseline (Image + Raw Question fusion). Performance stops improving after 5 guesses because the EVA-CLIP text encoder has a strict limit of 77 tokens.

As illustrated in Figure 3, injecting raw questions without the Entity Guesses mechanism (“No Guesses”) severely bounds both Hit Rate (25.8%) and Aggregate Score (27.6%). Conversely, the introduction of the hypotheses generation triggers an immediate performance surge, proving that our Entity Guessing mechanism successfully bridges the semantic gap with the image-only FAISS index.

However, performance stops improving and slightly degrades for $n \geq 10$. This occurs because forcing the model to generate a high number of names inevitably introduces inaccurate hypotheses, which adds semantic noise to the search query. Furthermore, if the number of guesses is even increased (such as $n = 20$), the text exceeds the 77-token limit of the EVA-CLIP encoder. At that point, the excess text is automatically truncated, making any extra words completely ineffective.

We selected **5 Guesses** as our definitive configuration. This value achieves the highest aggregate score (30.0%). Furthermore, parallel experiments at higher temperatures ($\tau = 0.3$) showed a consistent degradation in Hit Rate (averaging $\sim 26.8\%$), confirming that accurate hypothesis generation strictly requires low-temperature determinism to avoid semantic drift.

6 Final Results

6.1 Peak Performance on Qwen2.5-VL-3B

Table 2 reports the performance of our fully assembled agentic architecture ($k = 8$, $\alpha = 0.3$, 5 Guesses) on the 1000-sample test set, compared against our zero-shot baseline and the best static RAG configuration identified in Chapter 3.

Configuration	All	Hit rate	Score hit	Score miss
Plain VLM (Baseline)	17.6%	–	–	–
Static RAG (Entity Guessing, $k = 3$)	24.4%	26.3%	59.6%	12.0%
Our Agentic RAG ($\alpha = 0.3$, $k = 8$)	30.0%	28.3%	60.5%	18.1%

Table 2: Final results for the fully optimized Agentic pipeline. The integration of the ReAct loop, passages-level Critic filtering, and code-level fallback sets a peak performance for our project.

The system achieves an aggregate score of **30.0%**, exceeding our best Static RAG configuration by 5.6 percentage points. The results explicitly highlights the crucial role of the agentic loop and the Critic module.

In the Static RAG setup, despite achieving a solid Hit Rate (26.3%), the overall score was dragged down to 24.4%. This occurred because statically injecting un-filtered documents confused the model, as evidenced by the severe drop in accuracy when retrieval failed or introduced noise (**Score | miss** = 12.0%).

By transitioning to the Agentic RAG architecture (operating at $k = 8$ with 5 explicit guesses), the Hit Rate improves to **28.3%**. More importantly, the integration of the ReAG-Critic effectively shields the VLM from semantic noise, allowing the model to maintain a strong reading accuracy on correct documents (**Score | hit** = 60.5%). Furthermore, when the model answer with certain bailout patterns (like "Unknown" or "I don't know"), the programmatic Code-Level Fallback successfully prevents hallucinations, recovering plain-VLM reasoning (**Score | miss** = 18.1%, surpassing the plain baseline). This definitively demonstrates that a compact 3B model can successfully orchestrate a knowledge-grounded ReAct loop when the action space is constrained and context is filtered.

6.1.1 Autonomous Retrieval Bypass Analysis

A key feature of our ReAct-style architecture is the ability of the agent to autonomously bypass external retrieval via the `assess_retrieval_need` tool. To evaluate the effectiveness of this decision-making process, we isolated the subset of episodes where the 3B agent explicitly chose the `ANSWER_DIRECTLY` path, bypassing the vector store and the Critic.

Log analysis reveals that the agent autonomously skipped retrieval in **11.6% of the episodes** (116 out of 1000). On this specific subset, the zero-shot generation accuracy reached **26.7%**, significantly outperforming the global plain-VLM baseline (17.6%) and the *Score | miss* baseline (18.1%).

This result proves that the agent does not bypass retrieval at random; rather, it effectively estimates its own parametric uncertainty. By skipping retrieval only when the visual evidence is unambiguous, the agent successfully saves substantial computational budget (bypassing

both the FAISS search and the multi-passage ReAG-Critic evaluation) while capitalizing on high-confidence internal knowledge.

7 Conclusion

This project investigated a Multimodal Agentic RAG pipeline for the Encyclopedic-VQA benchmark, demonstrating that a compact 3B-parameter VLM can successfully orchestrate autonomous ReAct-style workflows through rigorous architectural design.

To address the image-to-image modality gap, we implemented a Entity Guessing strategy. Fusing generated text hypotheses with visual features proved strictly superior to pure visual retrieval, strongly aligning with the Identify-Before-Answer (IBA) framework [4]. Furthermore, we mitigated semantic noise by combining full-document retrieval with fine-grained passage filtering via the ReAG-Critic [3], effectively decoupling retrieval volume from prompt length.

Supported by a Code-Level Fallback mechanism to prevent hallucinations, our final autonomous pipeline achieved a peak score of **30.0%**, bypassing retrieval in high-confidence scenarios to optimize computational budget. Future work could explore integrating a purely textual retrieval tool to completely bypass the visual modality gap, fully leveraging the semantic precision of the generated hypotheses.

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